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Table 1.5 Range and Order of Lengths

Size of objects and distances	Length (m)
Distance to the boundary of observable universe	10^{26}
Distance to the Andromeda galaxy	10^{22}
Size of our galaxy	10^{21}
Distance from Earth to the nearest star (other than the Sun)	10^{16}
Average radius of Pluto's orbit	10^{12}
Distance of the Sun from the Earth	10^{11}
Distance of Moon from the Earth	10^{8}
Radius of the Earth	10^{7}
Height of the Mount Everest above sea level	10^{4}
Length of a football field	10^{2}
Thickness of a paper	10^{-4}
Diameter of a red blood cell	10^{-5}
Wavelength of light	10^{-7}
Length of typical virus	10^{-8}
Diameter of the hydrogen atom	10^{-10}
Size of atomic nucleus	10^{-14}
Diameter of a proton	10^{-15}

Some Common Practical Units

(i) Fermi = 1 fm = 10^{-15} m

(ii) 1 angstrom = 1 Å = 10^{-10} m

(iii) 1 nanometer = 1 nm = 10^{-9} m

(iv) 1 micron = 1μm = 10^{-6} m

(v) 1 Light year (Distance travelled by light in vacuum in one year) 1 Light Year = 9.467×10^{15} m

(vi) 1 astronomical unit (the mean distance of the Earth from the Sun) 1 AU = 1.496×10^{11} m

(vii) 1 parsec (Parallactic second) (Distance at which an arc of length 1 AU subtends an angle of 1 second of arc)

1 parsec = 3.08×10^{16} m =3.26 light year



Figure 1.7 The international 1 kg standard of mass, a platinum-iridium (9:1) cylinder 3.9 cm in height and diameter.



Why is the cylinder used in defining kilogram made up of platinum-iridium alloy?

This is because the platinum-iridium alloy is least affected by environment and time.

Chandrasekhar Limit (CSL) is the largest practical unit of mass.

1 CSL = 1.4 times the mass of the Sun

The smallest practical unit of time is Shake.

1 Shake = 10^{-8} s